DS ASSIGNMENT 1

- ANUSRI R

1.Sum of two matrices.

PROGRAM:

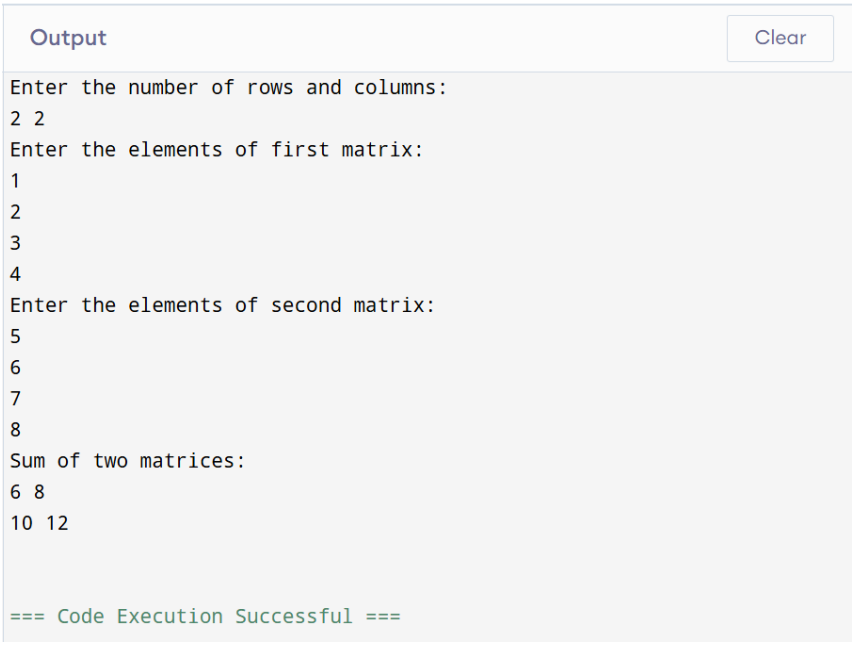
```
#include <stdio.h>
int main()
{
    int a[10][10], b[10][10], sum[10][10];
    int r, c;
    printf("Enter the number of rows and columns:\n");
    scanf("%d %d", &r, &c);
    printf("Enter the elements of first matrix:\n");
    for(int i = 0; i < r; i++)
    {
        for(int j = 0; j < c; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }
    printf("Enter the elements of second matrix:\n");
    for(int i = 0; i < r; i++)
    {
        for(int j = 0; j < c; j++)
        {
            scanf("%d", &b[i][j]);
        }
    }

    for(int i = 0; i < r; i++)
```

```
    {
        for(int j = 0; j < c; j++)
        {
            sum[i][j] = a[i][j] + b[i][j];
        }
    }
    printf("Sum of two matrices:\n");
    for(int i = 0; i < r; i++)
    {
        for(int j = 0; j < c; j++)
        {
            printf("%d ", sum[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT:



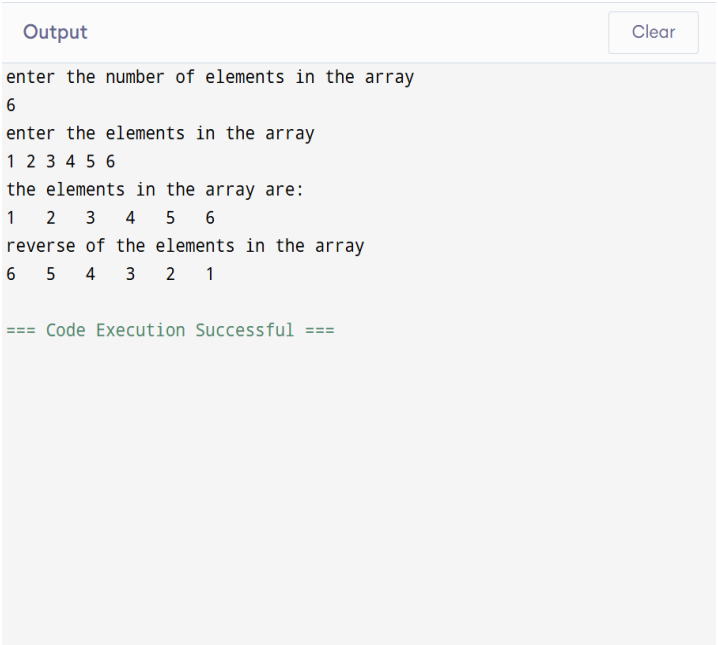
2. Write a C program to read n numbers into an array and display them in reverse order

PROGRAM:

```
# include <stdio.h>
int main()
{
int n,arr[20],reverse[20],j=0;
printf("enter the number of elements in the array\n");
scanf("%d",& n);
printf("enter the elements in the array\n");
for(int i=0;i<n;i++)
{
scanf("%d",&arr[i]);
    }
    printf("the elements in the array are:\n");
    for(int i=0;i<n;i++)
    {
        printf("%d\t",arr[i]);
```

```
    }
    printf("\n");
    for(int i=n-1;i>=0;i--)
      {
        reverse[j]=arr[i];
        j++;
      }
      printf("reverse of the elements in the array\n");
    for(int i=0;i<n;i++)
    {
        printf("%d\t",reverse[i]);
    }
  return 0;
}
```

OUTPUT:



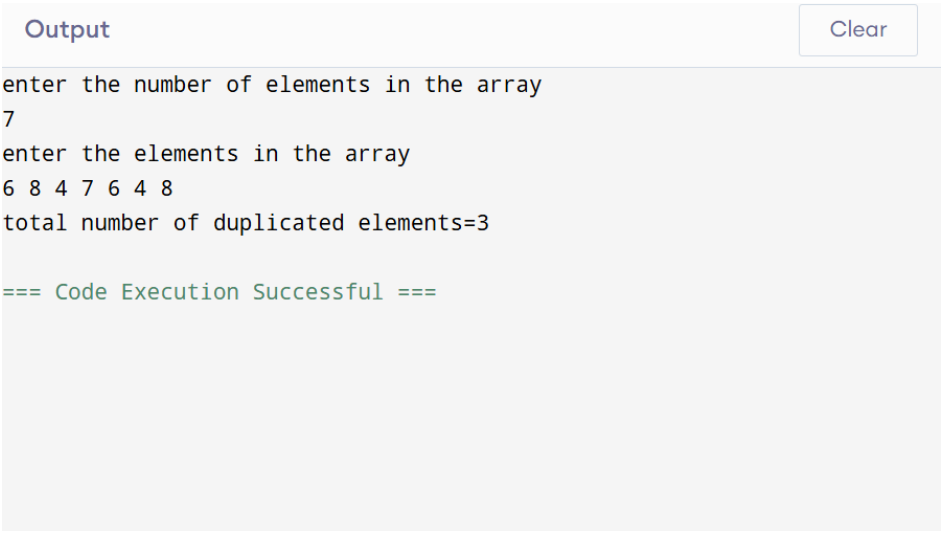
3. Count the total number of duplicate elements in an array.

PROGRAM:

```
# include <stdio.h>
int main()
{
    int arr[20],n;
    printf("enter the number of elements in the array\n");
    scanf("%d",& n);
    printf("enter the elements in the array\n");
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    int count=0;
    for(int j=0;j<n;j++)
    {
        int dup=0;
        int a=0;
        for(int k=0;k<j;k++)
        {
            if(arr[k]==arr[j])
            {
                a=1;
                break;
            }
        }
        if(a)
        {
            continue;
```

```
        }
    for(int i=j+1;i<n;i++)
    {
        if(arr[i]==arr[j])
        {
        dup=1;
        break;
        }
    }
    if(dup)
    {
        count++;
    }
}
printf("total number of duplicated elements=%d",count);
return 0;
}
```

OUTPUT:



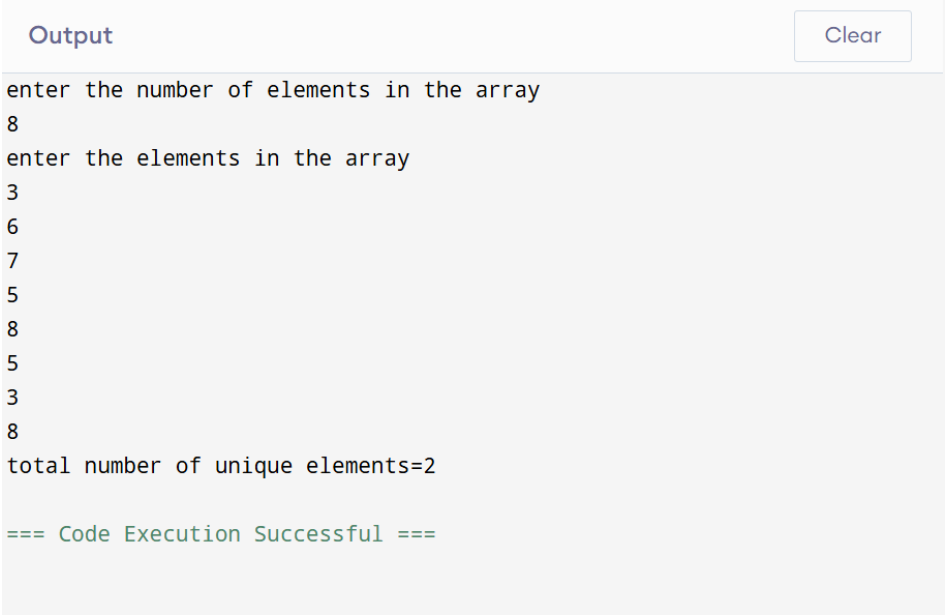
4.Print unique elements in an array.

PROGRAM:

```
# include <stdio.h>
int main()
{
    int arr[20],n;
    printf("enter the number of elements in the array\n");
    scanf("%d",& n);
    printf("enter the elements in the array\n");
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    int count=0;
    int unique=0;
    for(int j=0;j<n;j++)
    {
        int dup=0;
        int a=0;
        for(int k=0;k<j;k++)
        {
            if(arr[k]==arr[j])
            {
                a=1;
                break;
            }
        }
        if(a)
        {
```

```
            continue;
        }
    for(int i=j+1;i<n;i++)
    {
        if(arr[i]==arr[j])
        {
        dup=1;
        break;
        }
    }
    if(dup)
    {
        count++;
    }
    else
    {
        unique++;
    }
}
printf("total number of unique elements=%d",unique);
return 0;
}
```

OUTPUT:



5. Insert an element in a sorted array.

PROGRAM:

```
#include <stdio.h>
int main()
{
    int arr[20], n, val, ins = 0;

    printf("Enter the number of elements in the array:\n");
    scanf("%d", &n);
    printf("Enter the sorted elements:\n");
    for(int i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }
    printf("Enter the value to be inserted:\n");
    scanf("%d", &val);
    for(int i = 0; i < n; i++)
```

```
    {
        if(val < arr[i])
        {
            for(int j = n; j > i; j--)
            {
                arr[j] = arr[j - 1];
            }
            arr[i] = val;
            ins = 1;
            break;
        }
    }
    if(ins == 0)
    {
        arr[n] = val;
    }

    printf("After insertion:\n");

    for(int i = 0; i <= n; i++)
    {
        printf("%d ", arr[i]);
    }
    return 0;
}
```

OUTPUT:

Output Clear

```
Enter the number of elements in the array:
4
Enter the sorted elements:
3 4 5 6
Enter the value to be inserted:
2
After insertion:
2 3 4 5 6

=== Code Execution Successful ===
```

6. Insert an element in an unsorted array.

PROGRAM:

```
# include <stdio.h>
int main()
{
int arr[20],n,val,pos;
printf("enter the number of elements in the array\n");
    scanf("%d",&n);
    printf("enter the  elements\n");
    for(int i=0;i<n;i++)
    {
    scanf("%d",&arr[i]);
    }
    printf("enter the value to be inserted\n");
    scanf("%d",&val);
    printf("enter the position needed to insert\n");
    scanf("%d",&pos);
    for(int i=n;i>pos;i--)
    {
        arr[i]=arr[i-1];
```

```
    }
    arr[pos]=val;
    printf("after insertion\n");
    for(int i=0;i<=n;i++)
    {
        printf("%d\n",arr[i]);
    }
}
```

OUTPUT:

main.c | Output

```
enter the number of elements in the array
6
enter the  elements
2 4 7 54 67
78
enter the value to be inserted
38
enter the position needed to insert
3
after insertion
2
4
7
38
54
67
78


=== Code Execution Successful ===
```

7. Matrix multiplication.

PROGRAM:

```
#include <stdio.h>
```

```
int main()
{
    int r1, r2, c1, c2;
    int A[20][20], B[20][20], C[20][20];
    printf("Enter the rows and columns of the first matrix:\n");
    scanf("%d %d", &r1, &c1);
    printf("Enter the rows and columns of the second matrix:\n");
    scanf("%d %d", &r2, &c2);

    if(c1 != r2)
    {
        printf("Invalid, matrix multiplication can't be performed");
        return 0;
    }
    printf("Enter the data for first matrix:\n");
    for(int i = 0; i < r1; i++)
    {
        for(int j = 0; j < c1; j++)
        {
            scanf("%d", &A[i][j]);
        }
    }
    printf("Enter the data for second matrix:\n");
    for(int i = 0; i < r2; i++)
    {
        for(int j = 0; j < c2; j++)
        {
            scanf("%d", &B[i][j]);
        }
```

```
    }
    for(int i = 0; i < r1; i++)
    {
        for(int j = 0; j < c2; j++)
        {
            C[i][j] = 0;

            for(int k = 0; k < c1; k++)
            {
                C[i][j] += A[i][k] * B[k][j];
            }
        }
    }
    printf("Matrix multiplication result:\n");
    for(int i = 0; i < r1; i++)
    {
        for(int j = 0; j < c2; j++)
        {
            printf("%d ", C[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

Output:

main.c | Output

```
Enter the rows and columns of the first matrix:
2 2
Enter the rows and columns of the second matrix:
2 3
Enter the data for first matrix:
4 6
4 7
Enter the data for second matrix:
5 8 9
6 8 3
Matrix multiplication result:
56 80 54
62 88 57


=== Code Execution Successful ===
```

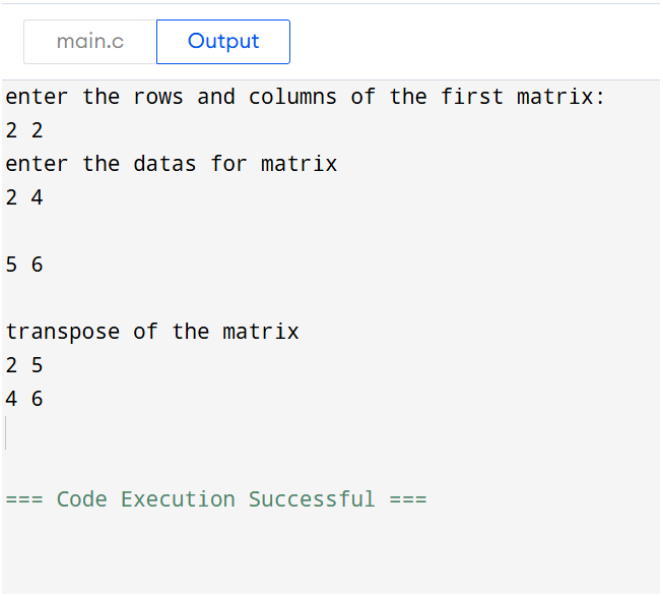
8. Matrix transpose.

PROGRAM:

```
# include <stdio.h>
int main()
{
int r1,r2,c1,c2,A[20][20];
printf("enter the rows and columns of the first matrix:\n");
scanf("%d %d",&r1,&c1);
printf("enter the datas for matrix\n");
for(int i=0;i<r1;i++)
{
{
}
for(int j=0;j<c1;j++)
scanf("%d",& A[i][j]);
printf("\n");
}
printf("transpose of the matrix\n");
```

```
for(int i=0;i<c1;i++)
{
{
}
for(int j=0;j<r1;j++)
printf("%d ",A[j][i]);
printf("\n");
}
return 0;
}
```

OUTPUT:



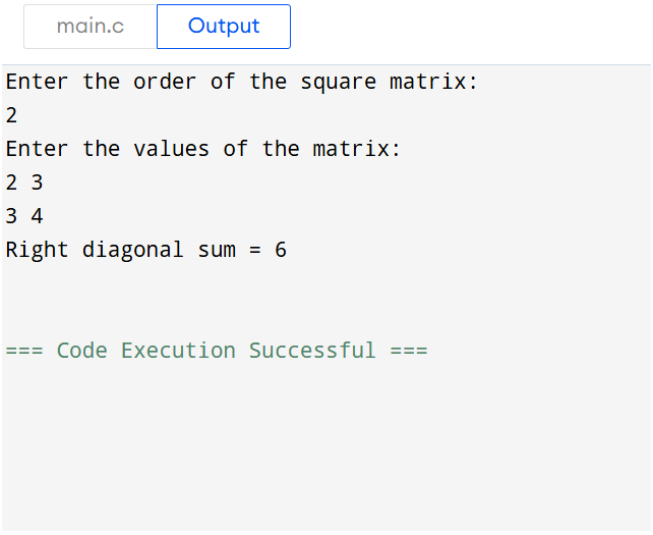
9.Right diagonal sum of a matrix

PROGRAM:

```
#include <stdio.h>
int main()
{
    int n, A[10][10], sum = 0;
    printf("Enter the order of the square matrix:\n");
```

```
    scanf("%d", &n);
    printf("Enter the values of the matrix:\n");
    for(int i = 0; i < n; i++)
    {
        for(int j = 0; j < n; j++)
        {
            scanf("%d", &A[i][j]);
        }
    }
    for(int i = 0; i < n; i++)
    {
        sum = sum + A[i][n - i - 1];
    }
    printf("Right diagonal sum = %d\n", sum);
    return 0;
}
```

OUTPUT:



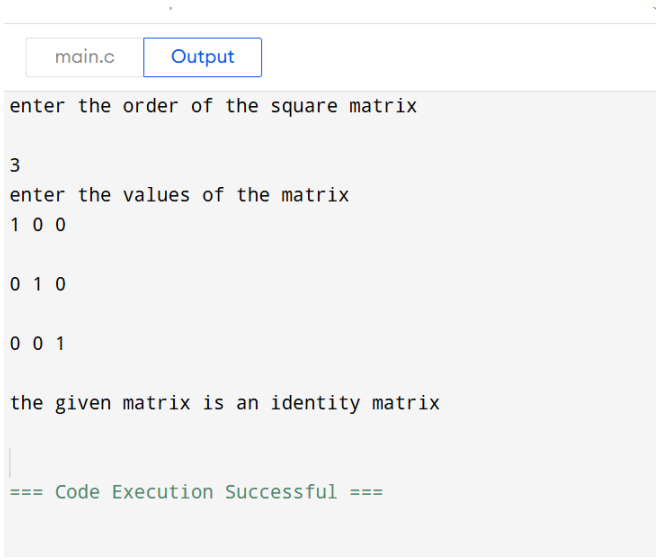
10. Check whether a matrix is an identity matrix.

PROGRAM:

```
# include <stdio.h>
int main()
{
int n,A[10][10],f=1;
printf("enter the order of the square matrix\n");
scanf("%d",&n);
printf("enter the values of the matrix\n");
for(int i=0;i<n;i++)
{
        for(int j=0;j<n;j++)
        {
            scanf("%d",&A[i][j]);
        }
        printf("\n");
    }
    for(int i=0;i<n;i++)
    {
        for(int j=0;j<n;j++)
        {
           if(i==j)
           {
            if(A[i][j]!=1)
            {
                f=0;
                break;
            }
           }
           else
```

```
        {
         if(A[i][j]!=0)
         {
            f=0;
            break;
         }
        }
      }
      if(f==0)
      break;
   }
   if(f==1)
   {
      printf("the given matrix is an identity matrix\n");
   }
   else{
      printf("the given matrix is not an identity matrix\n");
   }
   return 0;
}
```

OUTPUT:



11. Find the number of vowels, consonants, digits, and white spaces in a string.

PROGRAM:

```
# include <stdio.h>
int main()
{
    char str[50];
    printf("enter the string\n");
    fgets(str,50,stdin);
    int i=0,vow=0,con=0,space=0,num=0;
    while(str[i]!='\0')
    {
        if(('a'<=str[i] && 'z'>=str[i])||('A'<=str[i]&& 'Z'>=str[i]))
        {
        if(str[i]=='A'||str[i]=='E'||str[i]=='I'||str[i]=='O'||str[i]=='U'||str[i]=='a'||str[i]=='i'||str[i
]=='e'||str[i]=='o'||str[i]=='u')
        {
            vow++;
        }
```

```
    else
    {
        con++;
    }
}
if(str[i]==' ')
{
    space++;
}
if(str[i]>='0' && str[i]<='9')
{
    num++;
}
i++;
    }
    printf("number of vowels in the string=%d\n",vow);
    printf("number of consonents in the string=%d\n",con);
    printf("number of spaces in the string=%d\n",space);
    printf("number present in the string count =%d\n",num);
return 0;
}
```

OUTPUT:

main.c | Output

```
enter the string
HI I AM ANUSRI BORN ON 2008
number of vowels in the string=8
number of consonents in the string=9
number of spaces in the string=6
number present in the string count =4


=== Code Execution Successful ===
```

12. Find the frequency of a character in a string.

PROGRAM:

```
#include <stdio.h>
int main() {
    char str[100];
    int i, j, count;
    printf("Enter string: ");
    scanf("%s", str);
    for (i = 0; str[i] != '\0'; i++) {
        if (str[i] == '*')
            continue;
        count = 1;
        for (j = i + 1; str[j] != '\0'; j++) {
            if (str[i] == str[j]) {
                count++;
                str[j] = '*';
            }
        }
        printf("%c = %d\n", str[i], count);
```

```
    }
    return 0;
}
```

OUTPUT:

main.c | Output

```
Enter string: ANUSRI
A = 1
N = 1
U = 1
S = 1
R = 1
I = 1


=== Code Execution Successful ===
```

13. Count the number of words in a string.

PROGRAM:

```
# include <stdio.h>
int main()
{
    char str[50];
    printf("enter the string\n");
    fgets(str,50,stdin);
    int i=0,space=0;
    while(str[i]!='\0')
    {
        if(str[i]==' ')
        {
            space++;
        }
```

```
        i++;
    }
    printf("count of words in the string=%d",space+1);
    return 0;
}
```

OUTPUT:

main.c | Output

```
enter the string
hello guys how are you
count of words in the string=5

=== Code Execution Successful ===
```

14. Sort a string array.

PROGRAM:

```
# include <stdio.h>
# include <string.h>
int main()
{
    int n;
    char str[10][30],temp[30];
    printf("enter the number of strings:\n");
    scanf("%d",&n);
    getchar();
    printf("enter the strings:\n");
    for(int i=0;i<n;i++)
    {
       fgets(str[i],30,stdin);
       str[i][strcspn(str[i],"\n")]='\0';
```

```
    }
    for(int i=0;i<n-1;i++)
    {
        for(int j=i+1;j<n;j++)
        {
            if(strcmp(str[i],str[i+1])>0)
            {
              strcpy(temp,str[i]);
              strcpy(str[i],str[i+1]);
              strcpy(str[i+1],temp);
            }
        }
    }
    printf("sorted strings:\n");
    for(int i=0;i<n;i++)
    {
        printf("%s\n",str[i]);
    }
return 0;
}
```

OUTPUT:

main.c | Output

```
enter the number of strings:
5
enter the strings:
anu
jelin
press
kalai
kani
sorted strings:
anu
jelin
kalai
kani
press


=== Code Execution Successful ===
```

15. Toggle the case of a sentence (lowercase to uppercase and vice versa

PROGRAM:

```
# include <stdio.h>
int main()
{
    char str[100];
    printf("enter a sentence:\n");
    fgets(str,100,stdin);
    for(int i=0;str[i]!='\0';i++)
    {
        if(str[i]>='a' && str[i]<='z')
        {
            str[i]=str[i]-32;
        }
       else if(str[i]>='A' && str[i]<='Z')
        {
            str[i]=str[i]+32;
```

```
        }
    }
    printf("toggled sentence:\n");
printf("%s",str);
return 0;
}
```

OUTPUT:

main.c | Output

```
enter a sentence:
My Nattive is KATTATHUR. I Love my village.
toggled sentence:
mY nATTIVE IS kattathur. i lOVE MY VILLAGE.


=== Code Execution Successful ===
```